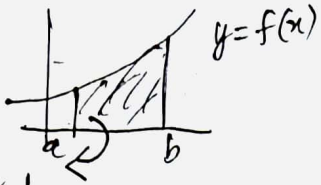
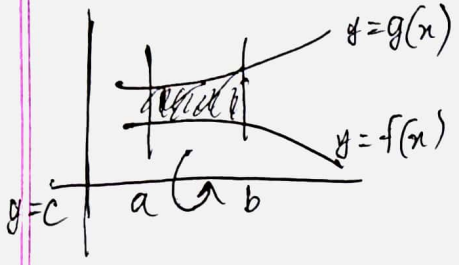
$$\text{Area} = \int_a^b y\,dx$$



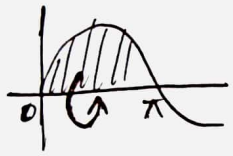
$$A = 2\pi \int_a^b |f(x)| \sqrt{1+(f'(x))^2}\,dx$$



$f(x) \le y \le g(x)$, $a \le x \le b$

$$V = \pi \int_a^b [(g(x)-c)^2 - (f(x)-c)^2]\,dx$$

Find the volume of revolution of the region bounded by $0 \le y \le \sin x$, $0 \le x \le \pi$, x-axis / $y=0$



$$V = \pi \int_0^\pi [(\sin x - 0)^2 - (0-0)^2]\,dx$$

$$= \pi \int_0^\pi \sin^2 x\,dx = \pi \times \frac{1}{2} \int_0^\pi (1-\cos 2x)\,dx = \frac{\pi}{2}\left[x - \frac{\sin 2x}{2}\right]_0^\pi$$

$$= \frac{\pi^2}{2}$$

$0 \le y \le \sin x$

$0 \le x \le \pi$

$y = -1$

$$V = \pi \int_0^\pi [(\sin x + 1)^2 - (0+1)^2]$$

$$= \pi \int_0^\pi \sin^2 x + 2\sin x + 1 - 1$$

$$= \pi \int_0^\pi \sin^2 x + 2\sin x \cdot = \pi \int_0^\pi \sin^2 x + 2\pi \int_0^\pi \sin x\,dx$$

$$= \frac{\pi^2}{2} + 2\pi[-\cos x]_0^\pi$$

$$= \frac{\pi^2}{2} + 2\pi[-\cos\pi - \cos 0]$$

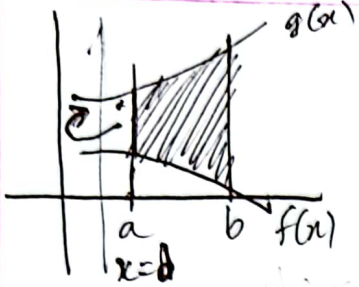
$$= \frac{\pi^2}{2} - 2\pi[\cos\pi - \cos 0]$$

$$= \frac{\pi^2}{2} - 2\pi(-1-1)$$

$$= \frac{\pi^2}{2} + 4\pi$$

x	1	2	3
x^2	1	4	9

x	1	2	3
$\sqrt{x}$	1	4	9



$a \le x \le b$

$f(x) \le y \le g(x)$

$x = d$

$$V = 2\pi \int_a^b (x-d)(g(x)-f(x))\,dx$$

$0 \le y \le \sin x$

$0 \le x \le \pi$

y-axis $(x=0)$

$$V = 2\pi \int_0^{\pi} (x-0)(\sin x - 0)\,dx$$

$$= 2\pi \int_0^{\pi} x \sin x\,dx$$

$$= 2\pi \left[x\int \sin x\,dx - \int \left(1 \int \sin x\,dx\right) dx \right]$$

$$= 2\pi \left[-x\cos x + \sin x \right]_0^{\pi}$$

$$= 2\pi \times \pi$$

$$= 2\pi^2$$

$$V = 2\pi \int_0^{\pi} (x+\pi)(\sin x - 0)\,dx$$

$$= 2\pi \int_0^{\pi} x \sin x\,dx + 2\pi^2 \int_0^{\pi} \sin x\,dx$$

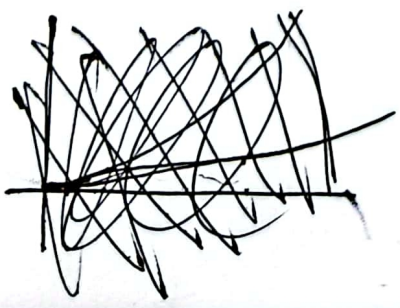
$$= 2\pi \cdot \pi + 2\pi^2 (-\cos x)_0^{\pi}$$

$$= 2\pi^2 + 2\pi^2 (-\cos\pi - (-\cos 0))$$

$y = x^2$, $y = \sqrt{x}$

Find the volume if the region enclosed by these graphs is revolved around a) X-axis, b) Y-axis

$y = x^2$



$\sqrt{x} = x^2$

$\sqrt{x} - x^2 = 0$

$\sqrt{x}(1 - x^{3/2}) = 0$

$\sqrt{x} = 0$ | $1 - x^{\frac{3}{2}} = 0$

$x = 0$ | $x = 1$

$V = \pi \int [(g(x) - 0)^2 - (f(x) - 0)^2] dx$ ($g(x) \to \sqrt{x}$, $f(x) \to x^2$)

$V = 2\pi \int_0^1 (x - 0)(g(x) - f(x)) dx$

$V = \pi \int [(\sqrt{x})^2 - (x^2)^2] dx$

$= 2\pi \int_0^1 (x - 0)(\sqrt{x} - x^2) dx$

111

```
class Testclass:
    name = 'testname'
    def __init__(self):
        self.a = 5
        self.b = 10
    @classmethod
    def clMethod(cls):
        print(cls.name)
```

Method — Instance, class, Static

```
    @staticmethod
    def stMethod():
        print('This is a static method')
    @classmethod
    def classMethod(cls, txt):
        print(cls.classname)
        print(txt)
```

Testclass.ClassMethod(cls 'Hello')

2k, 3, 4
1, 3, 2

Gamma Function

$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx \qquad \Gamma(n+1) = \int_0^\infty e^{-x} x^n dx$

Beta Function

$\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx$

$\beta(3,2) = \int_0^1 x^2(1-x)^1 dx = \int_0^1 x^2(1-x)\,dx = \int_0^1 x^2 - x^3\,dx = \left[\frac{x^3}{3}\right]_0^1 - \left[\frac{x^4}{4}\right]_0^1$

$= \frac{1}{3} - \frac{1}{4}$

$= \frac{1}{12}$

$\beta(2,3) = \int_0^1 x^{2-1}(1-x)^{3-1} dx$

$= \int_0^1 x(1-x)^2 dx$

$= \int_0^1 x(1-2x+x^2)\,dx$

$= \int_0^1 x - 2x^2 + x^3\,dx$

$= \left[\frac{x^2}{2}\right]_0^1 - \left[2\frac{x^3}{3}\right]_0^1 + \left[\frac{x^4}{4}\right]_0^1 dx$

$= \frac{1}{2} - \frac{2}{3} + \frac{1}{4}$

$= \frac{6-8+3}{12} = \frac{1}{12}$

$\beta(m,n) = \beta(n,m)$

$\beta(m,n) = \int_0^1 y^{m-1}(1-y)^{n-1} dy$

$y = 0 \to x = \infty$

$y = \frac{1}{1+x}$

$dy = \frac{1}{(1+x)^2} dx$

$= \int_\infty^0 \left(\frac{1}{1+x}\right)^{m-1} \left(1 - \frac{1}{1+x}\right)^{n-1} \left(-\frac{1}{(1+x)^2}\right) dx$

$= -\int_\infty^0 \frac{1}{(1+x)^{m-1}} \left(\frac{x}{1+x}\right)^{n-1} \cdot \frac{1}{(1+x)^2} dx$

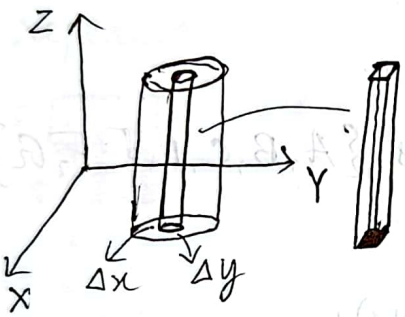
$= \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}$

$y = \sin x$

$$\beta(m,n) = 2\int_0^{\frac{\pi}{2}} \sin^{2m-1}x \cos^{2n-1}x \, dx$$

$$\beta(m,n) = \frac{\Gamma m \, \Gamma n}{\Gamma(m+n)}$$

Double Integral



$\iiint dx\,dy\,dz$

$\iint dA = dx\,dy$

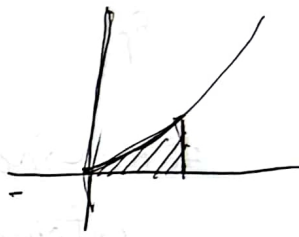
$$\int_{x=0}^{1}\int_{y=0}^{x^2} dy\,dx$$

$$= \int_{x=0}^{1} [y]_0^{x^2} dx$$

$$= \int_{x=0}^{1} x^2 dx$$

$$= \left[\frac{x^3}{3}\right]_0^1$$

$$= \frac{1}{3}$$



[y এর Limit x এর উপর Depend করেতা dy আগে করতে হবে]

Mass per unit area = xy

Mass = Mass per unit area × Area

$$= \iint xy \, dx\,dy$$

$$= \int_{x=0}^{1} x \int_{y=0}^{x^2} y \, dy \, dx$$

$$= \int_{x=0}^{1} x \left[\frac{y^2}{2}\right]_0^{x^2} dx = \frac{1}{2}\int_{x=0}^{1} x x^4 dx = \frac{1}{2}\left[\frac{x^6}{6}\right]_0^1$$

Multiple Integral

$y=f(x)$ → partial derivative এর opposite

$u=f(x,y,z)$

$\iint(x^2+3xy+y^2)\,dx\,dy$ → constant যেহেতু x এর সাপেক্ষে integration $\left[\frac{\partial}{\partial x}(y)=0\right]$

$=\int\left(\frac{x^3}{3}+3y\frac{x^2}{2}+y^2x\right)dy+f(y)$

$=\frac{x^3}{3}y+3\frac{x^2}{2}\frac{y^2}{2}+x\frac{y^3}{3}+f_1(y)+f_2(x)$

$\#\int_0^1 f(y)\,dy$ where $f(y)=\int_0^y y^2\sin xy\,dx$

$=\int_0^1\int_0^y y^2\sin xy\,dx\,dy$

$=\int_0^1\left[y^2\frac{\cos xy}{y}\right]_0^y dy$

$=\int_0^1\left[-y\cos xy\right]_0^y dy$

$y^2=u$

$\Rightarrow 2y\,dy=du$

$=\int_0^1(-y\cos y^2+y)\,dy$

$=\int_0^1 y-y\cos y^2\,dy$

$=\left[\frac{y^2}{2}\right]_0^1-\int\cos u\frac{du}{2}$

$=\left(\frac{1}{2}-0\right)-\frac{1}{2}\sin u$

$=\frac{1}{2}-\frac{1}{2}\left[\sin y^2\right]_0^1$

$=\frac{1}{2}-\frac{1}{2}(\sin 1-0)$

$=0.07926$

$1 \le y \le 4$, $xy = 4$, $2x = 5 + 4y - y^2 = 9 - (y-2)^2$

$\Rightarrow y - 2 = \sqrt{9 - 2x}$

$y = 2 \pm \sqrt{9 - 2x}$

$$\int_1^4 \int_{\frac{4}{2}}^{\frac{5+4y-y^2}{2}} dx\, dy$$

$$= \int_1^4 [x]_{\frac{4}{2}}^{\frac{5+4y-y^2}{2}} dy$$

$$= \int_1^4 \left[\frac{5+4y-y^2}{2} - \frac{4}{y}\right] dy$$

$$= \left[\frac{5}{2}y + \frac{4}{2}\frac{y^2}{2} - \frac{1}{2}\frac{y^3}{3} - 4\ln y\right]_1^4$$

$$= \left(\frac{5}{2}\cdot 4 + 4^2 - \frac{1}{6}\cdot 4^3 - 4\ln 4\right) - \left(\frac{5}{2}\cdot 1 + 1^2 - \frac{1}{6}1^3 - 4\ln 1\right)$$

$$= 9.788155 - 3.3333$$

$$= 6.454821$$

$\int_0^1 \int_x^{x^2} \int_{xy}^{x^2y^3} xy\, dz\, dy\, dx$

$$= \int_0^1 \int_x^{x^2} [xyz]_{xy}^{x^2y^3} dy\, dx$$

$$= \int_0^1 \int_x^{x^2} (xyx^2y^3 - xyxy)\, dy\, dx$$

$$= \int_0^1 \int_x^{x^2} (x^3y^4 - x^2y^2)\, dy\, dx$$

$$= \int_0^1 x^3 \left[\frac{x^3y^5}{5} - \frac{x^2y^3}{3}\right]_x^{x^2} dx$$

$$= \int_0^1 \left(\frac{x^3 \cdot x^{10}}{5} - \frac{x^2 \cdot x^6}{3}\right) - \left(\frac{x^3 \cdot x^5}{5} - \frac{x^2 \cdot x^3}{3}\right) dx$$

$$= \int_0^1 \left(\frac{x^{13}}{5} - \frac{x^8}{3} - \frac{x^8}{5} + \frac{x^5}{3}\right) dx$$

CamScanner

$$= \left[\frac{1}{5}\frac{x^{14}}{14} - \frac{1}{3}\frac{x^9}{9} - \frac{1}{5}\cdot\frac{x^9}{9} + \frac{1}{3}\frac{x^6}{6}\right]_0^1$$

$$= \frac{1}{70} - \frac{1}{27} - \frac{1}{45} + \frac{1}{18}$$

$$= \frac{2}{189}$$

Conclusion:

To conclude, urban farming is insufficient for food security as it's limited by space, has less variation of crops and risks biodiversity.

Suggestions

1. Utilizing vertical farming techniques to maximize space efficiency.
2. Exploring alternative crops that can thrive in urban environments.
3. Establishing urban green spaces and wildlife corridors to support biodiversity.

CamScanner

Urban Farming is the Most Effective Solution to Achieve Food Security

Imagine a world where you live in a city, yet you have go fre produce a few steps away from your front door. Urban farming is t that imagination into reality.

120

Differential Equation

$\frac{dy}{dx} = 5+x$

$e^x dy + e^y dx = 0$ } Ordinary DE

$\frac{\partial x}{\partial x} = 5+x$ Partial DE

$\left(\frac{d^2y}{dx^2}\right)^3 + 5\left(\frac{dy}{dx}\right)^3 + y = 0$ ⇒ Order 2, degree 3

↓ Higest derivative → H.d. 2,3 powep

$\left(\frac{d^4y}{dx^4}\right)^2 + 3\left(\frac{d^3y}{dx^3}\right)^7 + 4y = 0$

Order 4, degree 2

Check if satisfies

~~#~~ $y = c_1 \sin 2x + c_2 \cos 2x$ $y'' + 4y = 0$ ~~satisfies~~

$y' = 2c_1 \cos 2x - 2c_2 \sin 2x$

$y'' = -4c_1 \sin 2x - 4c_2 \cos 2x$

$y'' + 4y = -4c_1 \sin 2x - 4c_2 \cos 2x + 4(c_1 \sin 2x + c_2 \cos 2x) = 0$

First order, first degree

Separation of variables

Separable equation

$y' = \frac{x+1}{y^4+1}$

$\Rightarrow y^4+1 \frac{dy}{dx} = x+1$

$\Rightarrow (y^4+1)dy = (x+1)dx$

$\Rightarrow \frac{y^5}{5} + y + c_1 = \frac{x^2}{2} + x + c_2$ / $\frac{y^5}{5} + y = \frac{x^2}{2} + x + c$

$e^{2x}dx + e^{3y}dy = 0$

$\Rightarrow e^{2x}dx = -e^{3y}dy$

$\Rightarrow \frac{e^{2x}}{2} = -\frac{e^{3y}}{3} + c$

$\frac{dy}{dx} = e^{2x+3y} = e^{2x} \cdot e^{3y}$

$\int e^{-3y}dy = \int e^{2x}dx$

$\Rightarrow \frac{e^{-3y}}{-3} = \frac{e^{2x}}{2} + c$

Homogeneous equation

$f(tx, ty) = f(x, y)$. $y = xu$

$y' = \frac{2xy}{x^2-y^2}$ } first এ check করবো separable কিনা, নাহলে Homogenous

$f(x, y) = \frac{2xy}{x^2-y^2}$

$f(tx, ty) = \frac{2tx \cdot ty}{(tx)^2-(ty)^2} = \frac{2xyt^2}{t^2x^2-t^2y^2} = \frac{t^2 2xy}{t^2(x^2-y^2)} = \frac{2xy}{x^2-y^2}$

∴ Homogenous

$$y' = \frac{2xy}{x^2 - y^2}$$

Let,

$$y = xu$$

$$y' = x\frac{du}{dx} + u$$

$$\therefore x\frac{du}{dx} + u = \frac{2 \cdot x \cdot xu}{x^2 - (xu)^2} = \frac{2x^2u}{x^2 - x^2u^2} = \frac{2u}{1-u^2}$$

$$\Rightarrow x\frac{du}{dx} = \frac{2u}{1-u^2} - u = \frac{u+u^3}{1-u^2}$$

$$\int \frac{1-u^2}{u(1+u^2)}\,du = \int \frac{1}{x}\,dx$$

$$\frac{1+u^2-2u^2}{u(1+u^2)}$$

$$\int \left(\frac{1}{u} - \frac{2u}{1+u^2}\right)du = \int \frac{1}{x}\,dx$$

$$\ln u - \ln(1+u^2) = \ln x + c$$

$$\ln \frac{y}{x} - \ln\left(1 + \frac{y^2}{x^2}\right) = \ln x + c$$

~~Exa~~

~~$M(x,y)\,dx + N(x,y)\,dy = 0$~~

Exact Equation

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial g}{\partial x} = M$$

$$\frac{\partial g}{\partial y} = N$$

$Mdx + Ndy = 0$

Exact $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$g(x,y)$

$\frac{\partial g}{\partial x} = M$

$\frac{\partial g}{\partial y} = N$

$\frac{\partial g}{\partial x}dx + \frac{\partial g}{\partial y}dy = 0 \rightarrow dg = 0 \quad g = c$

$f = f(x,y)$ $x = x(s,t)$, $y = y(s,t)$

$x^2y + h(y)$

$x^2 + \cdot$

$2xy\,dx + (1+x^2)dy = 0$

$M = 2xy$, $N = 1+x^2$

$\frac{\partial M}{\partial y} = 2x = \frac{\partial N}{\partial x} = 2x$ $\therefore$ the eqn is exact } Exact eqn হলে check করতে হবে

N থেকেও করা যায়

$\frac{\partial g}{\partial x} = M = 2xy \Rightarrow \int \frac{\partial g}{\partial x}dx = \int 2xy\,dx \cdot \Rightarrow g = 2y\frac{x^2}{2} + h(y)$

$\frac{\partial g}{\partial y} = x^2 + \frac{\partial h}{\partial y}$

$1 + x^2 = x^2 + \frac{\partial h}{\partial y} \left[\frac{\partial g}{\partial y} = N\right]$

$\frac{\partial h}{\partial y} = 1$

$\int \frac{\partial h}{\partial y}dy = \int dy$

$h = y + c$

$g = x^2y + y + c$

$c_1 = x^2y + y + c$

$\boxed{x^2y + y = c_2 \Rightarrow (c_1 - c_2)} \rightarrow$ final eqn

* $x = 1, y = 3$ এরকম value দেয়া থাকলে final eqn এ বসাতে হবে.

$y' = \frac{2+ye^{xy}}{2y - xe^{xy}}$

$f(tx, ty) = \frac{2+tye^{txty}}{2ty - txe^{txty}}$ X

$\Rightarrow \frac{dy}{dx} = \frac{2+ye^{xy}}{2y - xe^{xy}}$

$\Rightarrow (2y - xe^{xy})dy = (2+ye^{xy})dx$

$\rightarrow (2y - xe^{xy})dx + (xe^{xy} - 2y)dy = 0$

$\frac{\partial M}{\partial y} = 2 - xye^{xy}$ $M = 2+ye^{xy}$ $N = xe^{xy} - 2y$

$\frac{\partial M}{\partial y} = 0 \cdot e^{xy} + y \cdot e^{xy} \cdot x + e^{xy}$

$\frac{\partial N}{\partial x} = xe^{xy} \cdot y + 0 + e^{xy}$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ $\therefore$ eqn is exact

$\frac{\partial g}{\partial x} = M$

$\int \frac{\partial g}{\partial x} dx = \int 2 + ye^{xy}$

$g = 2x + y\frac{e^{xy}}{y} + h(y)$

$g = 2x + e^{xy} + h(y)$

$\frac{\partial g}{\partial y} = xe^{xy} + \frac{\partial h}{\partial y}$

$\Rightarrow xe^{xy} - 2y = xe^{xy} + \frac{\partial h}{\partial y}$

$\therefore \frac{\partial h}{\partial y} = -2y$

$\Rightarrow \int \frac{\partial h}{\partial y} = -2\int y\,dy$

$\Rightarrow h = -2\frac{y^2}{2} + c$

$\Rightarrow h = -y^2$

$g = 2x + e^{xy} - y^2 + c$

$c_1 = 2x + e^{xy} - y^2 + c$

$2x + e^{xy} - y^2 = c_2$

120

Answer any 5

1. Double integral
2. Triple "
3. Linear differential equation
4. Exact/Non-exact de
5. ~~Homogenous/Non-Homogenous~~ higher order de
6. Modelling with de

Assignment 2 → 18 April (by email)

Quiz 2 → 16 April (online)

~~# $dx + \frac{dy}{dx} + 2xy^2 = 0$~~

$f(x,y) = 2xy^2$

$f(tx,ty) = 2tx(ty)^2$ ✗

$dy = 2xy^2 dx$

N M

$\frac{\partial M}{\partial y} = 4xy$ $\frac{\partial N}{\partial x} = x1$

$\Rightarrow \frac{dy}{dx} = -2xy^2$

$\Rightarrow y^{-2}dy = -2x\,dx$

$\Rightarrow \frac{y^{-2+1}}{-2+1} = -2\frac{x^2}{2} + c$

$\Rightarrow -\frac{1}{y} = -x^2 + c$

$\Rightarrow -\frac{1}{3} = -(1)^2 + c$

$\Rightarrow c = \frac{2}{3}$

, $y(1) = 3$

$\frac{dx}{dt} = kx$ $\quad t=0 \rightarrow x=A$

$\Rightarrow \frac{1}{kx}dx = dt$

$\Rightarrow \int \frac{dx}{x} = \int k\,dt$

$\Rightarrow \ln x = kt + c$

$\Rightarrow \ln A = k.0 + c$

$\Rightarrow c = \ln A$

$\ln x = kt + \ln A$

$\ln x - \ln A = kt$

$\ln \frac{x}{A} = kt$

$\frac{x}{A} = e^{kt}$

$\therefore x = Ae^{kt}$ Growth $\quad x = Ae^{-kt}$ Decay

#Half life = h years. Remaining 10%

$x = \frac{A}{2}, t = H$

$\frac{A}{2} = Ae^{-kH}$

$\ln\left(\frac{1}{2}\right) = e^{-kH}$

$\ln\left(\frac{1}{2}\right) = -kH$

$\Rightarrow k = \frac{H}{\ln(\frac{1}{2})} = H$

$\Rightarrow k = \frac{-\ln(\frac{1}{2})}{H}$

$= \frac{\ln(2)}{H}$

$x = 10/A = 0.1A$

$x = Ae^{-kt}$

$\Rightarrow 0.1A = Ae^{-\frac{\ln 2}{H}t}$

$\Rightarrow 0.1 = e^{-\frac{\ln 2}{H}t}$

$\Rightarrow \ln(0.1) = -\frac{\ln 2}{H}t$

$\Rightarrow t = -\frac{H\ln(0.1)}{\ln 2}$

$\Rightarrow t = 3.32H$

Linear Equation

$\frac{dy}{dx} + Py = Q$

$\frac{dy}{dx} + P(x)y = Q(x)$ } Linear Eqn in terms of y, y এর power 1

Integrating factor

$I.F. = e^{\int P(x)\,dx}$ (C add করবো না)

$\frac{x^2}{2} = u$

$\frac{1}{2} \cdot 2x\,dx = du$

$x\,dx = du$

$\frac{dy}{dx} - xy = -x$

$P = -x$, $Q = -x$

$I.F. = e^{\int P\,dx}$

$= e^{\int -x\,dx}$

$= e^{-\frac{x^2}{2}}$

$= e^{-\frac{x^2}{2}}$

$e^{-\frac{x^2}{2}} \frac{dy}{dx} - e^{-\frac{x^2}{2}} xy = -xe^{-\frac{x^2}{2}}$

$\frac{d}{dx}\left(y \cdot e^{-\frac{x^2}{2}}\right) = -xe^{-\frac{x^2}{2}}$

$\int d\left(ye^{-\frac{x^2}{2}}\right) = -\int xe^{-\frac{x^2}{2}}\,dx$

$ye^{-\frac{x^2}{2}} = -\int e^{-u}\,du$

$= e^{-u} + c$

$= e^{-\frac{x^2}{2}} + c$

$y = 1 + ce^{\frac{x^2}{2}}$

$y' - 3y = 6$

$\Rightarrow y' = 6 + 3y$

$P = 3$, $Q = 6$

$I.F. = e^{\int -3\,dx}$

$= e^{-3x}$

$e^{-3x}y' - 3e^{-3x}y = 6e^{-3x}$

$\frac{d}{dx}\left(y \cdot e^{-3x}\right) = 6e^{-3x}$

$\Rightarrow ye^{-3x} = \int 6e^{-3x}\,dx$

$\Rightarrow ye^{-3x} = 6 \cdot \frac{e^{-3x}}{-3} + c$

$\Rightarrow ye^{-3x} = -2e^{-3x} + c$

$(-1)\ e^0 = -2e^0 + c$ ~~y^2~~ $(y(0) = -1)$

$c = 1$

$ye^{-3x} = -2e^{-3x} + 1$

$y = -2 + e^{3x}$ (যেহেতু val করলে particular solution, or else general sol

$$\frac{dI}{dt} + \frac{R}{L} I = \frac{E}{L}$$

$$\frac{dq}{dt} + \frac{q}{RC} = \frac{E}{L}$$

RL circuit

$E = 3\sin 2t$, $R = 10\,\Omega$, $L = 0.5\,H$, $I(0) = 6A$

$$\frac{dI}{dt} + \frac{10}{0.5} I = \frac{3\sin 2t}{0.5} \Bigg\} P = 20,\ Q = 6\sin 2t$$

$$\frac{dI}{dt} + 20I = 6\sin 2t$$

$$I.F. = e^{\int 20\,dt} = e^{20t}$$

$$e^{20t}\frac{dI}{dt} + 20I e^{20t} = 6e^{20t}\sin 2t$$

[exact না হলে যেটা দিয়ে গুণ করে exact হবে সেটাই I.F.]

$$e^{at} = \frac{e^{at}}{a}$$

$$\frac{d}{dt}(Ie^{20t})$$

$$Ie^{20t} = 6\int \underset{\downarrow v}{e^{20t}} \underset{\downarrow u}{\sin 2t}\,dt$$

$$= 6\left(\sin 2t \int e^{20t} - \int \frac{d}{dx}(\sin 2t)\left(e^{20t}\right)dt\right)$$

$$= 6\sin 2t \frac{e^{20t}}{20} - \text{'2 বাকি}$$

101

1 evidence

Annotate → main idea, evidence, margin notes

Detail about 1 supporting idea

$(x^2+y^2+x)dx + xy\,dy = 0$

$M = x^2+y^2+x$

$N = xy$

$\frac{\partial M}{\partial y} = 2y \qquad \frac{\partial N}{\partial x} = y$

$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{1}{xy}(2y-y) = \frac{1}{x} = f(x)$

$I.F. = e^{\int f(x)dx} = e^{\int \frac{1}{x}dx} = e^{\ln x} = x$

~~x^2~~

$\# (x^3+xy^2+x^2)dx + x^2y\,dy = 0$

$M = x^3+xy^2+x^2 \quad , \quad N = x^2y$

$\frac{\partial M}{\partial y} = 2xy \qquad \frac{\partial N}{\partial x} = 2xy$

$\frac{\partial g}{\partial x} = M = x^3+xy^2+x^2 \qquad N =$

$\Rightarrow g = \frac{x^4}{4} + \frac{x^2y^2}{2} + \frac{x^3}{3} + h(y)$

$h'(y) = 0$

$h(y) = c_1$

$g = \frac{x^4}{4} + \frac{x^2y^2}{2} + \frac{x^3}{3} + c_1$

Higher Order DE:

$\frac{dy}{dx} + xy = 2x$

- Separable
- Homogenous
- Exact
- Non-Exact → Integrating Factor
  I.F. $= e^{\int P dx}$
- Linear

$\frac{d^2y}{dx^2} + A_1\frac{dy}{dx} + A_2 y = 0$

$y = e^{mx}$

$\frac{dy}{dx} = me^{mx}$

$\frac{d^2y}{dx^2} = m^2e^{mx}$

$m^2e^{mx} + A_1 me^{mx} + A_2 e^{mx} = 0$

$\Rightarrow (m^2 + A_1 m + A_2)e^{mx} = 0$

$\Rightarrow m^2 + A_1 m + A_2 = 0 \longrightarrow$ Auxiliary Equation

$m = m_1, m_2$

$\frac{d^3y}{dx^3} \rightarrow m_1, m_2, m_3 \ldots$ $\quad \frac{d^ny}{dx^n} \rightarrow m = m_1, m_2, \ldots m_n$

$y = c_1e^{m_1x} + c_2e^{m_2x} + \ldots + c_ne^{m_nx}$

$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$

$y = e^{mx}$, $\frac{dy}{dx} = me^{mx}$, $\frac{d^2y}{dx^2} = m^2e^{mx}$

$m^2e^{mx} - 5me^{mx} + 6e^{mx} = 0$

$(m^2 - 5m + 6)e^{mx} = 0$ → $-\infty$ না হলে $\neq 0$ হবে না, $[e^{mx} \neq 0]$

$m^2 - 5m + 6 = 0$

$m = 3, 2$

$y = c_1e^{3x} + c_2e^{2x}$

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$

$y = e^{mx}$, $\frac{dy}{dx} = me^{mx}$, $\frac{d^2y}{dx^2} = m^2e^{mx}$

$m^2e^{mx} - 4me^{mx} + 4y \cdot e^{mx} = 0$

$\Rightarrow e^{mx}(m^2 - 4m + 4) = 0$

$\Rightarrow m^2 - 4m + 4 = 0$

$\Rightarrow (m-2)^2 = 0$

$\Rightarrow m = 2, 2$

$y = c_1e^{2x} + x c_2e^{2x}$ [solution equal হলে x^n add হবে, $n = 0, 1, 2, 3, \ldots$]

Auxiliary Equation $\rightarrow$ $m = m_1, m_2, \ldots, m_n$

$m^2 + pm + q = 0$

$m = m_1, m_2$

Roots are unequal : $y = c_1e^{m_1x} + c_2e^{m_2x}$ / $y = c_1e^{m_1x} + c_2e^{m_2x} + c_3e^{m_3x}$

Roots are equal : $y = c_1e^{m_1x} + xc_2e^{m_1x}$

$y = c_1e^{mx} + xc_2e^{mx} + x^3c_3e^{mx}$ [3 solutions]

$\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + y = 0$

$m^3 + 3m^2 + 3m + 1 = 0$

$(m+1)^3 = 0$

$m = -1, -1, -1$

$y = c_1e^{-x} + c_2xe^{-x} + c_3x^2e^{-x}$

$(m-1)(m-2)^2 = 0$

$m = 1, 2, 2$ $\rightarrow$ equal so এখানে x আসবে

$y = c_1e^{x} + c_2e^{2x} + xc_3e^{2x}$

$\frac{dy}{dx} + P(x)y = Q(x)$

$I.F. = e^{\int P(x)dx} = e^{-\frac{x^2}{2}}$

$\frac{dy}{dx} e^{-\frac{x^2}{2}} - xye^{-\frac{x^2}{2}} = xe^{-\frac{x^2}{2}}$

$\frac{d}{dx}\left(ye^{-\frac{x^2}{2}}\right) = xe^{-\frac{x^2}{2}}$

$ye^{-\frac{x^2}{2}} = \int xe^{-\frac{x^2}{2}} dx$

$= \int -du = -u + c$

$ye^{-\frac{x^2}{2}} = -e^{-\frac{x^2}{2}} + c$

$e^{-\frac{x^2}{2}} = u$

$\Rightarrow e^{-\frac{x^2}{2}} \cdot \left(\frac{-1}{2}\right) \cdot 2x\,dx = du$

$\Rightarrow -e^{-\frac{x^2}{2}} x\,dx = du$

$\Rightarrow$

$\frac{dI}{dt} + \frac{R}{L} I = \frac{E}{L}$ $\quad [I = y]$

Linear Equation

$$\frac{dy}{dx} + Py = Q \quad , \quad I.F. = e^{\int P\,dx}$$

$$\frac{dy}{dx} + y\tan x = \sec x$$

~~I.F. = e^{\int}~~

$$P = \tan x$$

$$Q = \sec x$$

$$I.F. = e^{\int \tan x\,dx} = e^{\ln \sec x} = \sec x$$

$$\sec x \cdot \frac{dy}{dx} + y \cdot \tan x \sec x = \sec^2 x$$

$$\frac{d}{dx}(y \sec x) = \sec^2 x$$

$$y \sec x = \int \sec^2 x\,dx = \tan x + c$$

$$y \sec x = \tan x + c$$

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$$I.F. = \frac{1}{x^2y^2}$$

$$\frac{1}{x^2y^2}\left((x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy\right) = 0$$

$$\left(\frac{1}{y} - \frac{2}{x}\right)dx - \left(\frac{x}{y^2} - \frac{3}{y}\right)dy = 0$$

$$M = \frac{1}{y} - \frac{2}{x}$$

$$N = \frac{3}{y} - \frac{x}{y^2}$$

$$\frac{\partial M}{\partial y} = -\frac{1}{y^2}$$

$$\frac{\partial N}{\partial x} = -\frac{1}{y^2}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$ exact

then the same procedure for exact DE.

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$$(x^2+y^2+x)\,dx + xy\,dy = 0$$

$M = x^2+y^2+x$

$N = xy$

$\frac{\partial M}{\partial y} = 2y$, $\frac{\partial N}{\partial x} = y$, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{1}{xy}(2y-y) = \frac{1}{x} = f(x)$

$I.F. = e^{\int f(x)dx} = e^{\int \frac{1}{x}dx} = e^{\ln x} = x$

$(x^3+xy^2+x^2)\,dx + x^2y\,dy = 0$

$M = x^3+xy^2+x^2$, $N = x^2y$

$\frac{\partial M}{\partial y} = 2xy$, $\frac{\partial N}{\partial x} = 2xy$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ ∴ exact

$\frac{\partial g}{\partial x} = M = x^3+xy^2+x^2$

or, $g = \int (x^3+xy^2+x^2)\,dx$

$g = \frac{x^4}{4} + y^2\cdot\frac{x^2}{2} + \frac{x^3}{3} + h(y)$

$\frac{\partial g}{\partial y} = x^2y + h'(y)$

$N = x^2y + h'(y)$

$x^2y = x^2y + h'(y)$

$$(1-xy)\,y\,dx - (1+xy)\,x\,dy = 0$$

$$M = (1-xy)\,y = y - xy^2$$

$$N = -(1+xy)\,x = -x - x^2y$$

$$\frac{\partial M}{\partial y} = 1 - 2xy$$

$$\frac{\partial N}{\partial x} = -1 - 2xy$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$\text{I.F.} = \frac{1}{Mx - Ny} = \frac{1}{xy - x^2y^2 + xy + x^2y^2}$$

$$= \frac{1}{2xy}$$

$$\frac{1}{2xy}\left((1-xy)\,y\,dx - (1+xy)\,x\,dy\right) = 0$$

$$\left(\frac{1}{2x} - \frac{y}{2}\right)dx - \left(\frac{1}{2y} + \frac{x}{2}\right)dy = 0$$

$$M = \frac{1}{2x} - \frac{y}{2}, \quad N = -\left(\frac{1}{2y} + \frac{x}{2}\right)$$

$$\frac{\partial M}{\partial y} = -\frac{1}{2}, \qquad \frac{\partial N}{\partial x} = -\frac{1}{2}$$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so exact.

- Reducible to exact

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

- $\dfrac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x) \rightarrow I.F. = e^{\int f(x)\,dx}$

- $\dfrac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} = f(y) \rightarrow I.F. = e^{-\int f(y)\,dy}$

- Homogeneous $\rightarrow f(tx, ty) = f(x, y)$

$$I.F. = \frac{1}{Mx + Ny}$$

- [redacted] =
- $M = y f(x,y)$, $N = x\, g(x,y)$

$$I.F. = \frac{1}{Mx - Ny}$$

Integrating Factor

$I\,(Mdx + Ndy) = 0$ is exact

$I(x,y)\,(M(x,y)dx + N(x,y)dx) = 0$ is exact

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$$h'(y) = 0$$

$$h(y) = c_1$$

$$g = \frac{x^4}{4} + \frac{x^2y^2}{2} + \frac{x^3}{3} + c_1$$

$$\frac{x^4}{4} + \frac{x^2y^2}{2} + \frac{x^3}{3} = c$$

- $(x^2y - 2xy^2)\,dx - (x^3 - 3x^2y)\,dy = 0$

$$M = x^2y - 2xy^2 \quad , \quad N = 3x^2y - x^3$$

$$\frac{\partial M}{\partial y} = x^2 - 4xy \quad , \quad \frac{\partial N}{\partial x} = 6xy - 3x^2$$

- $$\frac{dy}{dx} = \frac{x^2y - 2xy^2}{x^3 - 3x^2y} = f(x,y)$$

$$f(tx, ty) = \frac{(tx)^2\,ty - 2\cdot tx\,(ty)^2}{(tx)^3 - 3(tx)^2(ty)}$$

$$= \frac{t^3(x^2y - 2xy^2)}{t^3(x^3 - 3x^2y)}$$

$$= f(x,y).$$

Homogeneous,

$$I.F. = \frac{1}{Mx + Ny} = \frac{1}{x^3y - 2x^2y^2 + 3x^2y^2 - x^3y}$$

Differential Equations

Separable $\int A\,dy = \int x\,dx$

Homogeneous $f(tx, ty) = f(x, y)$

$y = xv \rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$x = yv \rightarrow \frac{dx}{dy} = v + y\frac{dv}{dy}$

Exact

$$M\,dx + N\,dy = 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$f(x, y)$

$\frac{\partial f}{\partial x} = M$ & $\frac{\partial f}{\partial y} = N$

$\frac{\partial f}{\partial x} = M$

$f = \int M\,dx$

$f = \ldots + h(y)$

$\frac{\partial f}{\partial y} = \ldots + h'(y) = N$

$\frac{\partial f}{\partial y} = N$

$f = \int N\,dy = \ldots + g(x)$

$\frac{\partial f}{\partial x} = \ldots + g'(x) = M$

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